Vectori și val. proprii. Diagonalizare

$Av = \lambda v$

vectori proprii = vector care nu își schimbă direcția

$$P_A(x) = \det(x I_n - A) = (-1)^n \det(A - x I_n)$$

Spectrul lui $A = \sigma(A) =$ mulț. val. proprii

1 Folosind H-C, calculați A^{-1}; A^4

$$A = \begin{pmatrix} 0 & 2 & 1 \\ -1 & 1 & 0 \\ 0 & 0 & 2 \end{pmatrix}$$

$$\det(A - x I_3) = 0 \Rightarrow \begin{vmatrix} -x & 2 & 1 \\ -1 & 1-x & 0 \\ 0 & 0 & 2-x \end{vmatrix} = 0 \Rightarrow$$

$$\Rightarrow -x(1-x)(2-x) + 2(2-x) = 0 \Rightarrow (2-x)[-x+x^2+2] = 0$$

$$\Rightarrow -2x + 2x^2 + 4 + x^2 - x^3 - 2x = 0 \Rightarrow -x^3 + 3x^2 - 4x + 4 = 0$$

I $2 - x = 0 \Rightarrow x = 2$

II $x^2 - x + 2 = 0$

$\Delta = 1 - 4\cdot 1 \cdot 2 < 0$ $\nexists x$

$\Rightarrow \sigma(A) = 2$

$$-A^3 + 3A^2 - 4A + 4I_3 = 0 \Rightarrow \frac{1}{4}(A^3 - 3A^2 + 4A) = I_3$$

$$A\,\underbrace{\frac{1}{4}(A^2 - 3A + 4I_3)}_{A^{-1}} = I_3$$

$$A^2 = \begin{pmatrix} 0 & 2 & 1 \\ -1 & 1 & 0 \\ 0 & 0 & 2 \end{pmatrix}\begin{pmatrix} 0 & 2 & 1 \\ -1 & 1 & 0 \\ 0 & 0 & 2 \end{pmatrix} = \begin{pmatrix} -2 & 2 & 2 \\ -1 & -1 & -1 \\ 0 & 0 & 4 \end{pmatrix}$$

$$A^{-1}=\frac{1}{4}(A^2-3A+4I_3)=\frac{1}{4}\left[\begin{pmatrix}-2&2&2\\-1&-1&-1\\0&0&4\end{pmatrix}-\begin{pmatrix}0&6&3\\-3&3&0\\0&0&6\end{pmatrix}+\begin{pmatrix}4&0&0\\0&4&0\\0&0&4\end{pmatrix}\right]$$

$$A^{-1}=\frac{1}{4}\begin{pmatrix}2&-4&-1\\2&0&-1\\0&0&2\end{pmatrix}=\begin{pmatrix}1/2&-1&-1/4\\1/2&0&-1/4\\0&0&1/2\end{pmatrix}$$

$$A^3=3A^2-4A+4I_3 \,\big|\cdot A \Rightarrow A^4=3A^3-4A^2+4A$$

$$A^4=3(3A^2-4A+4I_3)-4A^2+4A$$

$$A^4=9A^2-12A+12I_3-4A^2+4A$$

$$A^4=5A^2-8A+12I_3$$

$$A^4=5\begin{pmatrix}-2&2&2\\-1&-1&-1\\0&0&4\end{pmatrix}-8\begin{pmatrix}0&2&1\\-1&1&0\\0&0&2\end{pmatrix}+12I_3$$

$$A^4=\begin{pmatrix}2&-6&2\\3&-1&-5\\0&0&16\end{pmatrix}$$

2) Verificați dacă matricea A este diagonalizabilă, în caz afirmativ diagonalizați-o.

a) $A=\begin{pmatrix}1&2\\2&4\end{pmatrix}$

$$\det(A-\lambda I_2)=0 \Longrightarrow \begin{vmatrix}1-\lambda&2\\2&4-\lambda\end{vmatrix}=0 \Rightarrow (1-\lambda)(4-\lambda)-4=0$$

$$\cancel{4}-\lambda-4\lambda+\lambda^2-\cancel{4}=0 \Rightarrow \lambda^2-5\lambda=0 \Rightarrow \lambda(\lambda-5)=0$$

$$\Rightarrow \lambda_{1,2}\in\{0;5\}$$

$\boxed{\lambda_1=0}$

$$Av_1=\lambda_1 v_1 \Rightarrow \begin{pmatrix}1&2\\2&4\end{pmatrix}\begin{pmatrix}a\\b\end{pmatrix}=\begin{pmatrix}\lambda_1 a\\\lambda_1 b\end{pmatrix}=\begin{pmatrix}0\\0\end{pmatrix} \Rightarrow \begin{cases}a+2b=0\\2a+4b=0\end{cases}$$

Notăm $a = \alpha, \alpha \in \mathbb{R} \Rightarrow b = -\frac{\alpha}{2} \Rightarrow$

$$\Rightarrow v_1 = \begin{pmatrix} \alpha \\ -\alpha/2 \end{pmatrix} \Rightarrow v_1 = \alpha \begin{pmatrix} 1 \\ -1/2 \end{pmatrix}, \alpha \in \mathbb{R}$$

$\boxed{\lambda_2 = 5}$

$$Av_2 = \lambda_2 v_2 \Rightarrow \begin{pmatrix} 1 & 2 \\ 2 & 4 \end{pmatrix} \begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} \lambda_2 a \\ \lambda_2 b \end{pmatrix} = \begin{pmatrix} 5a \\ 5b \end{pmatrix} \Rightarrow \begin{cases} a + 2b = 5a \\ 2a + 4b = 5b \end{cases}$$

$$\Rightarrow \begin{cases} 2b = 4a \\ 2a = b \end{cases} \Rightarrow \text{Notăm } a = \alpha, \alpha \in \mathbb{R} \Rightarrow b = 2\alpha \Rightarrow$$

$$\Rightarrow v_2 = \begin{pmatrix} \alpha \\ 2\alpha \end{pmatrix} = \alpha \begin{pmatrix} 1 \\ 2 \end{pmatrix}, \alpha \in \mathbb{R}$$

A diagonalizabilă $\Leftrightarrow a(\lambda_i) = g(\lambda_i)$

$\lambda_1 = 0 \Rightarrow a(\lambda_1) = 1 \qquad \dim_K V_{\lambda_1} = 1 = g(\lambda_1)$

$\lambda_2 = 5 \Rightarrow a(\lambda_2) = 1 \qquad \dim_K V_{\lambda_2} = 1 = g(\lambda_2)$

$$\left.\begin{matrix} a(\lambda_1) = g(\lambda_1) \\ a(\lambda_2) = g(\lambda_2) \end{matrix}\right| \Rightarrow \text{A diagonalizabilă}$$

$$A = P \cdot \Delta \cdot P^{-1}$$

$$\Delta = \begin{pmatrix} 0 & 0 \\ 0 & 5 \end{pmatrix} \qquad P = \begin{pmatrix} -2 & 1 \\ 1 & 2 \end{pmatrix}$$

b) $A = \begin{pmatrix} -1 & 0 & 1 \\ 3 & 0 & -3 \\ 1 & 0 & -1 \end{pmatrix}$

$$\det(A - \lambda I_3) = 0 \Rightarrow \begin{vmatrix} -1-\lambda & 0 & 1 \\ 3 & -\lambda & -3 \\ 1 & 0 & -1-\lambda \end{vmatrix} = 0 \Rightarrow$$

$$\Rightarrow -\lambda(-1-\lambda)(-1-\lambda) + \lambda = 0 \Rightarrow -\lambda(1+\lambda)^2 + \lambda = 0 \Rightarrow$$

$$\Rightarrow \lambda[\cancel{1} - (\cancel{1} + 2\lambda + \lambda^2)] = 0 \Rightarrow -\lambda^2(2+\lambda) = 0 \Rightarrow \lambda_{1,2} \in \{0; -2\}$$

cu $a(\lambda_1) = a(0) = 2$

$a(\lambda_2) = a(-2) = 1$

$\boxed{\lambda_1=0}$

$$A v_1=\lambda_1 v_1 \Rightarrow \begin{pmatrix} -1 & 0 & 1 \\ 3 & 0 & -3 \\ 1 & 0 & -1 \end{pmatrix}\begin{pmatrix} a \\ b \\ c \end{pmatrix}=\begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} \Rightarrow \begin{cases} -a+c=0 \\ 3a-3c=0 \\ a-c=0 \end{cases} \Rightarrow$$

$$\Rightarrow a=c$$

Notăm $a=\alpha, \alpha\in\mathbb{R}$ $\Rightarrow V_{\lambda_1}=\left\{\begin{pmatrix} \alpha \\ \beta \\ \alpha \end{pmatrix} \middle| \alpha,\beta\in\mathbb{R}\right\}$

$y=\beta, \beta\in\mathbb{R}$

$$V_{\lambda_1}=\left\{\begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}\alpha+\begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}\beta \middle| \alpha,\beta\in\mathbb{R}\right\}$$

$$\Rightarrow V_{\lambda_1}=Sp\left\{\begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix};\begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}\right\} \quad \dim V_{\lambda_1}=2=g(\lambda_1)$$

$\boxed{\lambda_2=-2}$

$$A\cdot v_2=\lambda_2 v_2 \Rightarrow \begin{pmatrix} -1 & 0 & 1 \\ 3 & 0 & -3 \\ 1 & 0 & -1 \end{pmatrix}\begin{pmatrix} a \\ b \\ c \end{pmatrix}=\begin{pmatrix} -2a \\ -2b \\ -2c \end{pmatrix} \Rightarrow \begin{cases} -a+c=-2a \\ 3a-3c=-2b \\ a-c=-2c \end{cases}$$

$$\Rightarrow \begin{cases} c=-a \Rightarrow a=-c \\ -2b=-3c-3c \Rightarrow b=3c \end{cases}$$

Notăm $a=\alpha, \alpha\in\mathbb{R} \Rightarrow V_{\lambda_2}=\left\{\begin{pmatrix} \alpha \\ -3\alpha \\ -\alpha \end{pmatrix} \middle| \alpha\in\mathbb{R}\right\}=Sp\left\{\begin{pmatrix} 1 \\ -3 \\ -1 \end{pmatrix}\right\}$

$$\dim V_{\lambda_2}=1 \Rightarrow g(\lambda_2)=1$$

$$\left.\begin{array}{l} a(\lambda_1)=2=g(\lambda_1) \\ a(\lambda_2)=1=g(\lambda_2) \end{array}\right\} \Rightarrow A \text{ diagonalizabilă}$$

$$D=\begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & -2 \end{pmatrix} \qquad A=P\cdot D\cdot P^{-1}$$

$$P=\begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & -3 \\ 1 & 0 & -1 \end{pmatrix}$$

c) $A = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 2 & -5 & 4 \end{pmatrix}$

$$\det(A-\lambda I_3) = \begin{vmatrix} -\lambda & 1 & 0 \\ 0 & -\lambda & 1 \\ 2 & -5 & 4-\lambda \end{vmatrix} = \lambda^2(4-\lambda)+2-5\lambda$$

$$= 4\lambda^2-\lambda^3+2-5\lambda = 0 \Rightarrow \lambda^3-4\lambda^2+5\lambda-2=0$$

	λ^3	λ^2	λ	λ^0
	1	-4	5	-2
1	1	-3	2	0

$\Rightarrow \lambda_1 = 1$

$$\Delta = 9-4\cdot 1\cdot 2 = 1 \Rightarrow \lambda_{2,3} = \frac{3\pm 1}{2\cdot 1} \begin{cases} \lambda_2 = 2 \\ \lambda_3 = 1 \end{cases}$$

$\boxed{\lambda_1 = 1}$ $a(\lambda_1) = 2$

$$Av_1 = \lambda_1 v_1 \Rightarrow \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 2 & -5 & 4 \end{pmatrix}\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} x \\ y \\ z \end{pmatrix} \Rightarrow \begin{cases} x = y \\ z = y \\ 2x-5y+4z = z \end{cases}$$

$$2y-5y+4y = z \Rightarrow y = z$$

Notăm $x = \alpha, \alpha \in \mathbb{R} \Rightarrow V_{\lambda_1} = \left\{ \begin{pmatrix} \alpha \\ \alpha \\ \alpha \end{pmatrix} \middle| \alpha \in \mathbb{R} \right\} = Sp\left\{ \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} \right\}$

$\dim V_{\lambda_1} = 1 \Rightarrow g(\lambda_1) = 1$

$a(\lambda_1) \neq g(\lambda_1) \Rightarrow A$ nu e diagonalizabilă

3) Fie $A = \begin{pmatrix} 0 & 1 \\ 2 & 1 \end{pmatrix}$ Calculați A^{2024}

$$\det(A-\lambda I_2) = 0 \Rightarrow \begin{vmatrix} -\lambda & 1 \\ 2 & 1-\lambda \end{vmatrix} = 0 \Rightarrow -\lambda(1-\lambda)-2=0$$

$$\Rightarrow -\lambda+\lambda^2-2=0 \Rightarrow \lambda^2-\lambda-2=0$$

$$\Delta = 1+4\cdot 1\cdot 2 = 9 \Rightarrow \lambda_{1,2} = \frac{1\pm 3}{2\cdot 1} \begin{cases} \lambda_1 = -1 \\ \lambda_2 = 2 \end{cases}$$

cu $a(\lambda_1) = a(\lambda_2) = 1$

$\boxed{\lambda_1 = -1}$

$$\begin{pmatrix} 0 & 1 \\ 2 & 1 \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} -x \\ -y \end{pmatrix} \Rightarrow \begin{cases} y = -x \\ 2x+y = -y \end{cases} \Rightarrow x = -y$$

$$V_{\lambda_1} = \left\{ \begin{pmatrix} -\alpha \\ \alpha \end{pmatrix} \middle| \alpha \in \mathbb{R} \right\} = Sp\left\{ \begin{pmatrix} -1 \\ 1 \end{pmatrix} \right\} \Rightarrow \dim_{\mathbb{K}} V_{\lambda_1} = 1$$

$$g(\lambda_1) = 1$$

$\boxed{\lambda_2 = 2}$

$$\begin{pmatrix} 0 & 1 \\ 2 & 1 \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 2x \\ 2y \end{pmatrix} \Rightarrow \begin{cases} y = 2x \\ 2x+y = 2y \end{cases} \Rightarrow \begin{cases} y = 2x \\ 2x = y \end{cases} \Rightarrow y = 2x$$

$$V_{\lambda_1} = \left\{ \begin{pmatrix} \alpha \\ 2\alpha \end{pmatrix} \middle| \alpha \in \mathbb{R} \right\} = Sp\left\{ \begin{pmatrix} 1 \\ 2 \end{pmatrix} \right\} \Rightarrow \dim_{\mathbb{K}} V_{\lambda_2} = 1$$

$$g(\lambda_2) = 1$$

$$\left.\begin{array}{l} a(\lambda_1) = g(\lambda_1) = 1 \\ a(\lambda_2) = g(\lambda_2) = 1 \end{array}\right| \Rightarrow A \text{ diagonalizabilă} \Rightarrow D = \begin{pmatrix} -1 & 0 \\ 0 & 2 \end{pmatrix}$$

$$P = \begin{pmatrix} 1 & 1 \\ -1 & 2 \end{pmatrix}$$

$$\det P = 2-(-1) = 3 \Rightarrow P^{-1} = \frac{1}{3}\begin{pmatrix} 2 & -1 \\ 1 & 1 \end{pmatrix}$$

$$A = P\cdot D\cdot P^{-1} \Rightarrow A^{2024} = P\cdot D^{2024}\cdot P^{-1}$$

$$A^{2024} = \begin{pmatrix} 1 & 1 \\ -1 & 2 \end{pmatrix}\begin{pmatrix} (-1)^{2024} & 0 \\ 0 & 2^{2024} \end{pmatrix}\frac{1}{3}\begin{pmatrix} 2 & -1 \\ 1 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 2^{2024} \\ -1 & 2^{2025} \end{pmatrix}\begin{pmatrix} 2/3 & -1/3 \\ 1/3 & 1/3 \end{pmatrix}$$

$$A^{2024} = \begin{pmatrix} \frac{2+2^{2024}}{3} & \frac{-1+2^{2024}}{3} \\ \frac{-2+2^{2025}}{3} & \frac{1+2^{2025}}{3} \end{pmatrix}$$

④ Fie $f: \mathbb{R}^3 \to \mathbb{R}^3$ endomorfism unde $M(f) = \begin{pmatrix} 5 & -3 & 2 \\ 6 & -4 & 4 \\ 4 & -4 & 5 \end{pmatrix}$

Arătați că f e diagonalizabil

$\det(M(f) - \lambda I_3) = 0 \Rightarrow \begin{vmatrix} 5-\lambda & -3 & 2 \\ 6 & -4-\lambda & 4 \\ 4 & -4 & 5-\lambda \end{vmatrix} = 0 \Rightarrow$

$\Rightarrow (5-\lambda)^2(-4-\lambda) - 3\cdot 16 - 8\cdot 6 + 8(4+\lambda) + 16(5-\lambda) + 18(5-\lambda) = 0$

$(25 - 10\lambda + \lambda^2)(-4-\lambda) - 96 + 32 + 8\lambda + 80 - 16\lambda + 90 - 18\lambda = 0$

$-100 - 25\lambda + 40\lambda + 10\lambda^2 - 4\lambda^2 - \lambda^3 + 106 - 26\lambda = 0$

$-\lambda^3 + 6\lambda^2 - 11\lambda + 6 = 0$

	λ^3	λ^2	λ	λ^0
	1	-6	11	-6
1	1	-5	6	0
2	1	-3	0	

$\lambda_1 = 1$
$\lambda_2 = 2$
$\lambda_3 = 3$
$\Rightarrow$ val. proprii reale; distinc. $\Rightarrow f$ diag.

Val. proprii sunt reale și distincte $\Rightarrow$

$\Rightarrow A$ diagonal.